Question 1

What is the exact value of $\cos \frac{x}{2}$, given the information that $\tan x = \frac{8}{15}, 180^\circ \leq x \leq 270^\circ$

A) $\frac{1}{\sqrt{17}}$

B) $-\frac{1}{\sqrt{17}}$

C) $\frac{4}{\sqrt{17}}$

D) $-\frac{4}{\sqrt{17}}$

Question 2

Which of the following is the solution set of the equation $\sin 2x = \cos x$ on the interval $[0, 2\pi)$

A. $\{\frac{\pi}{6}, \frac{5\pi}{6}\}$

B. $\{\frac{\pi}{6}, \frac{5\pi}{6}, \frac{8\pi}{6}, \frac{3\pi}{2}\}$

C. $\{\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}\}$

D. $\{\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}\}$

E. $\{\frac{\pi}{6}, \frac{\pi}{2}, \frac{7\pi}{6}, \frac{3\pi}{2}\}$

Question 3

A ball on a spring is pulled 6 inches below its rest position and then released. After t seconds, the ball's distance, d, in inches from its rest position is given by $d = -6 \cos \frac{\pi t}{2}$. For which of the following values of t, the ball is 3 inches below its rest position.

- 3/2
- 5/3
- 1/3
- 10/3